Schrodinger numerical with different potentials

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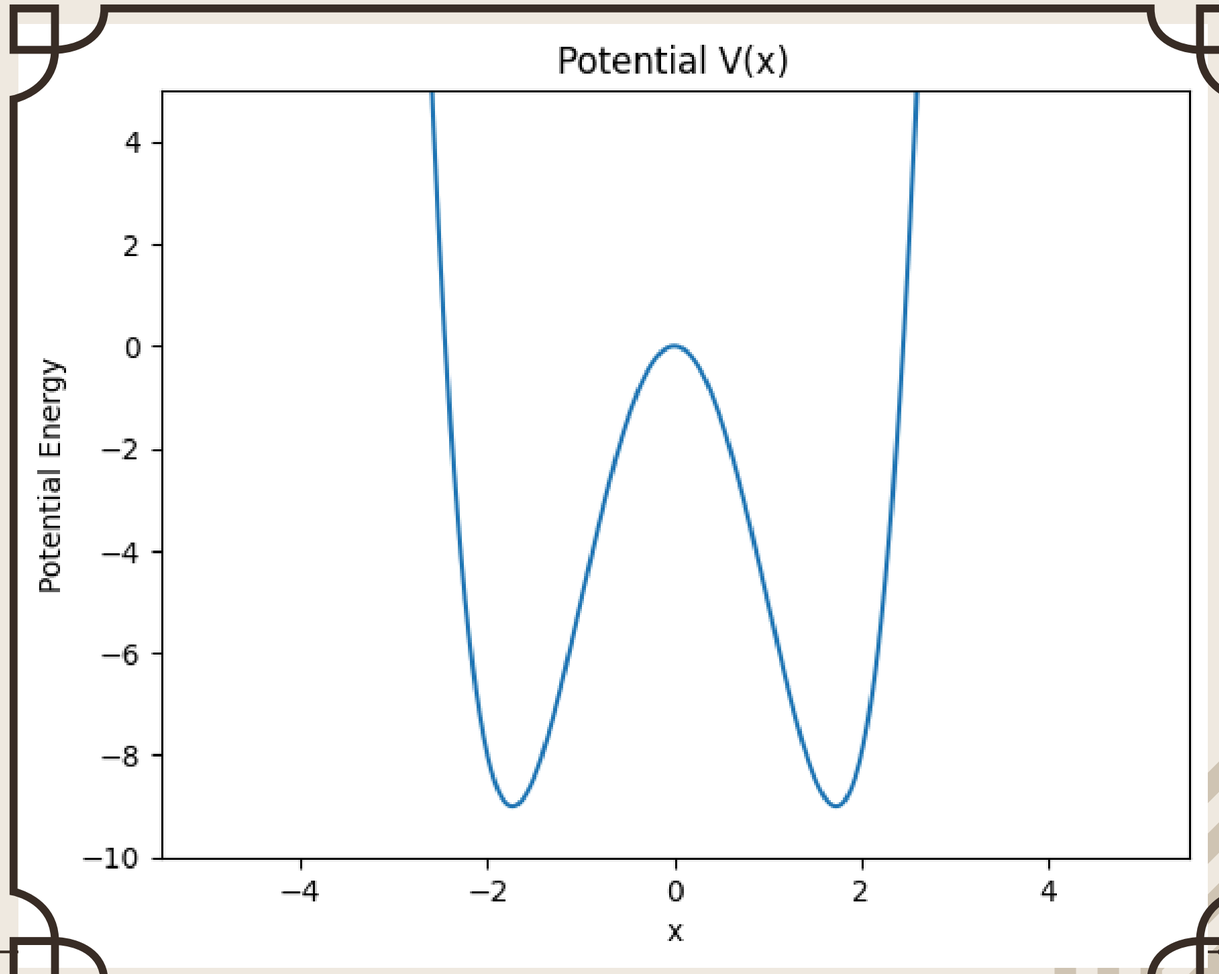
Double-well potential

$V(X) = B \cdot X^4 - C \cdot X^2$,
AMB $B = 1$ I $C = 6$.

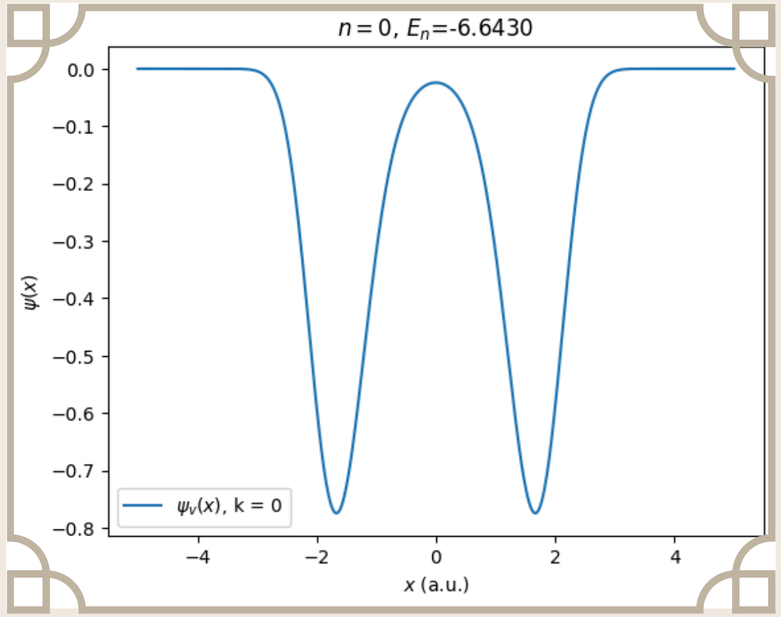
SUBSTITUINT ELS VALORS:

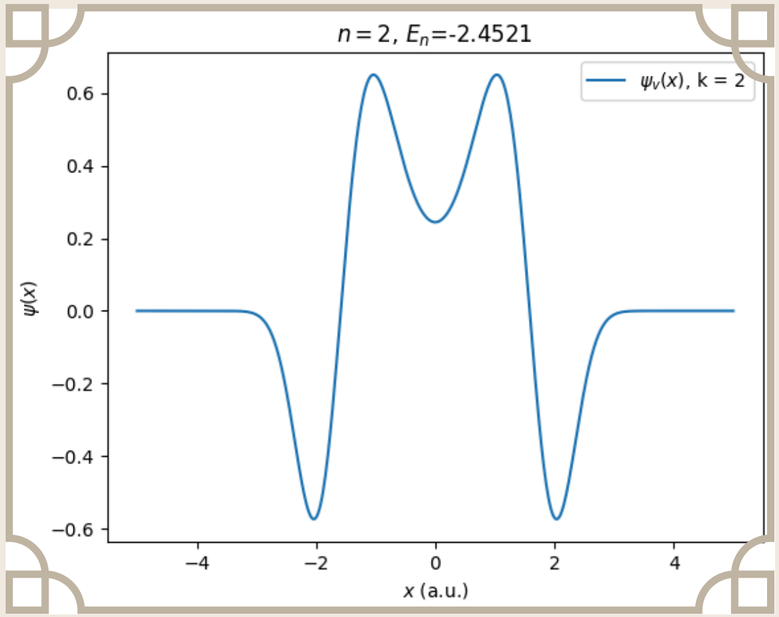
$V(X) = X^4 - 6X^2$.
$H = 0.02$

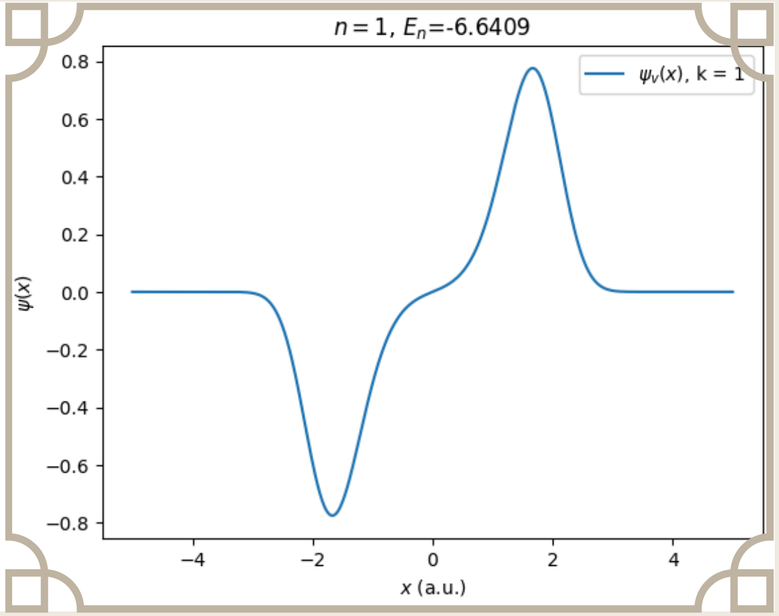


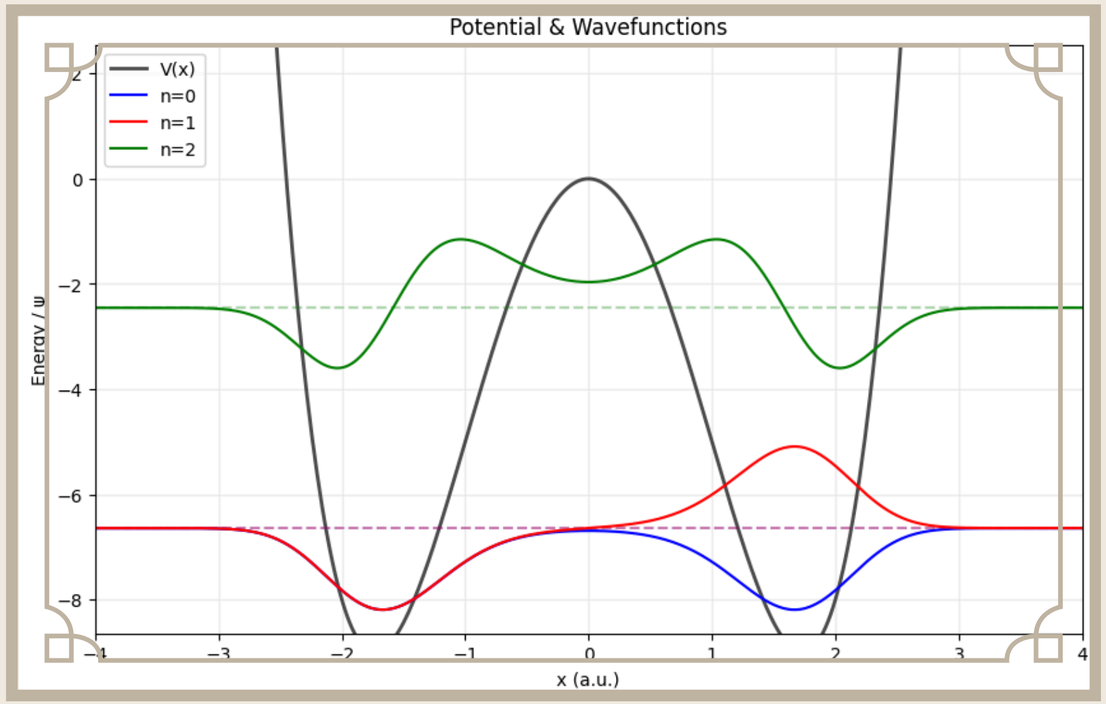
H=0,02
E(0)=-6.64299
E(1)=-6.64089
E(2)=-2.45211
E(3)=-2.31664
E(4)=0.41427

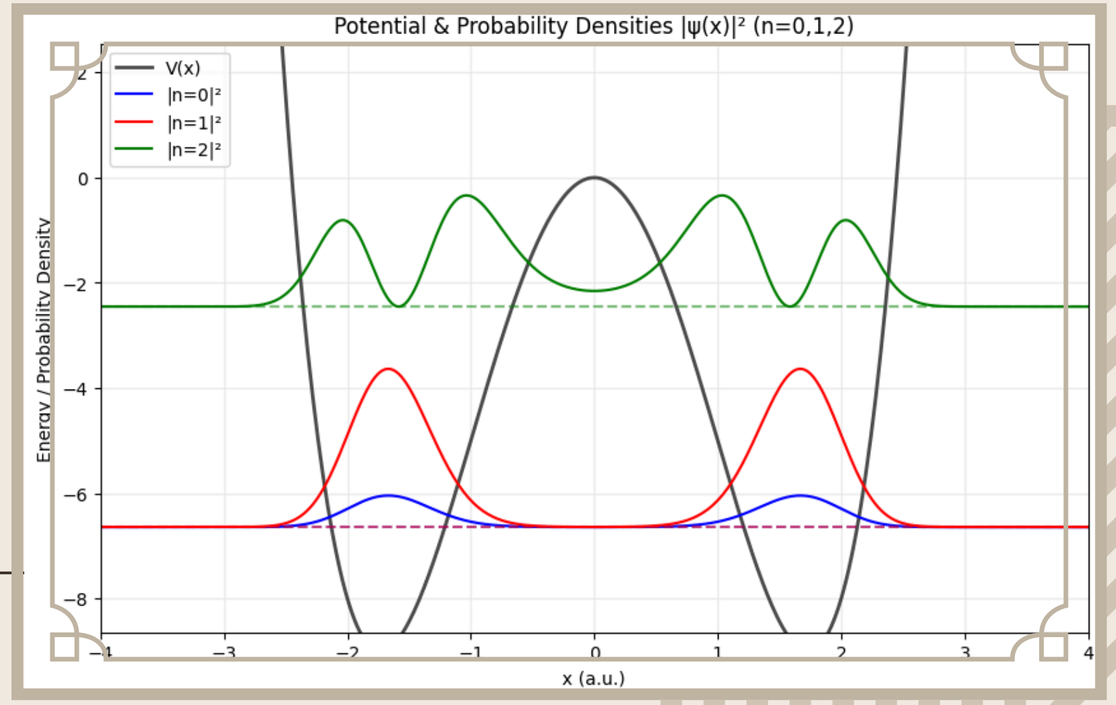
H=0,1
E(0)=-6.64944
E(1)=-6.64731
E(2)=-2.47450
E(3)=-2.34248
E(4)=0.38188



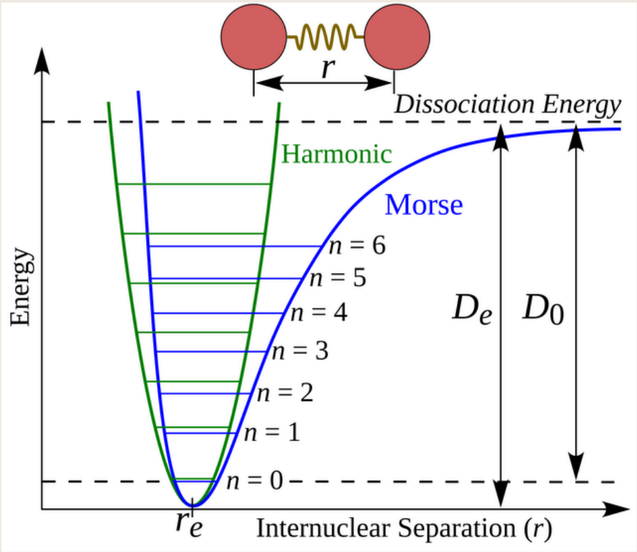





Potential & Wavefunctions
V(x)
n=0
n=1
n=2
Energy / ω
x (a.u.)


Potential & Probability Densities |ψ(x)|² (n=0,1,2)
V(x)
|n=0|²
|n=1|²
|n=2|²
Energy / Probability Density
x (a.u.)

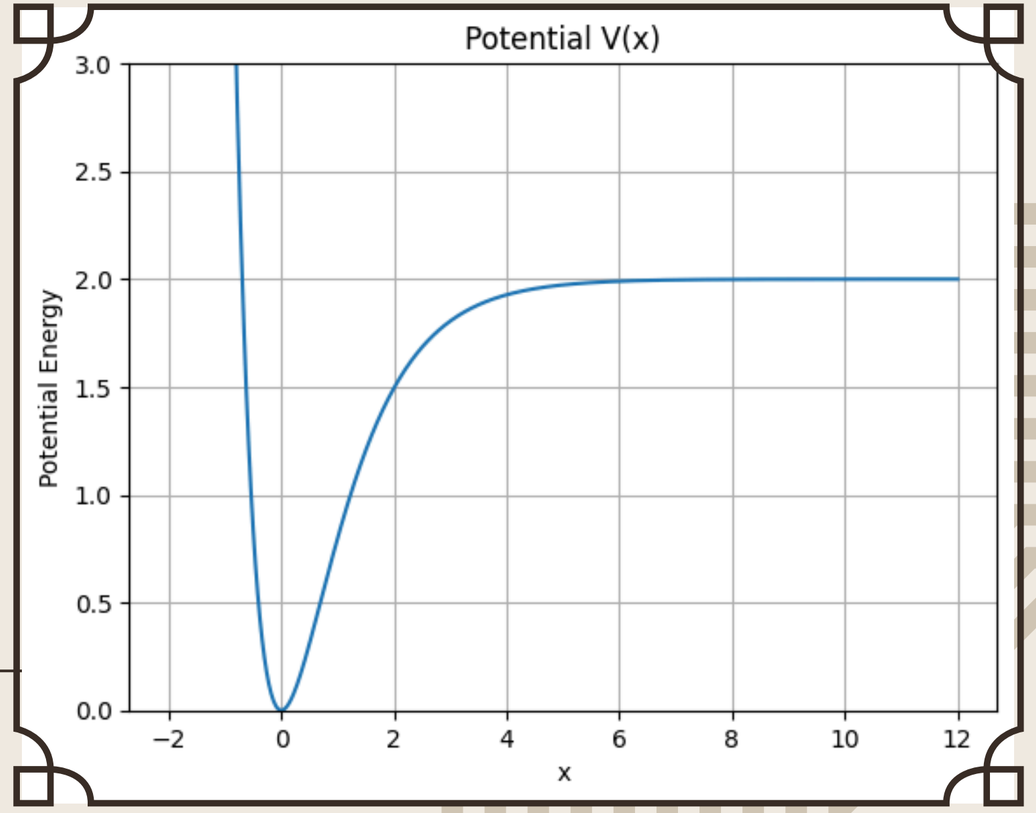
Morse potential

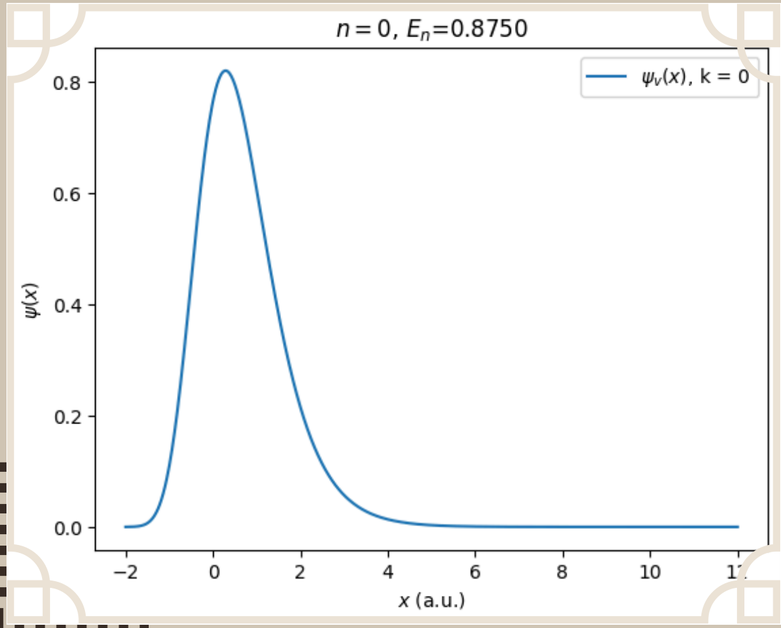


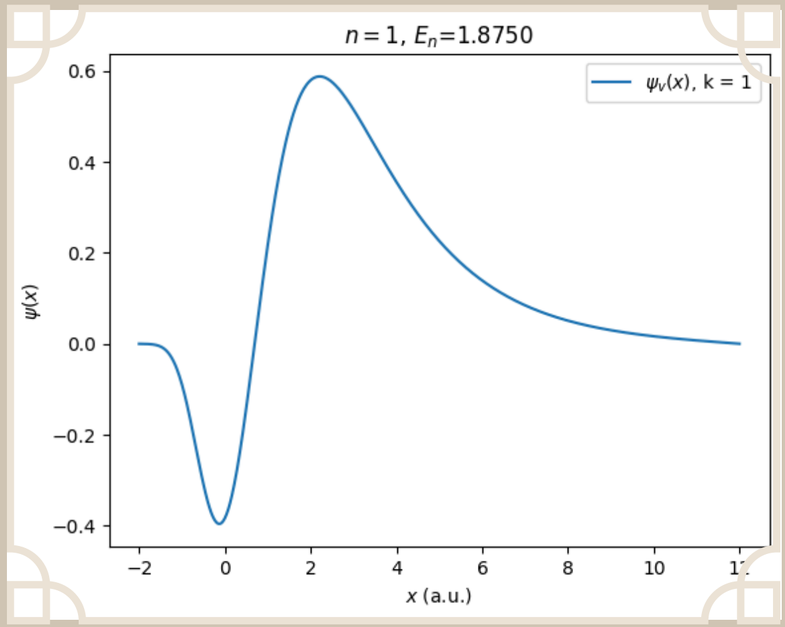
$V(X) = D \cdot [1 - EXP(-B \cdot (X - X_e))]^2$, AMB $D = 2$, $B = 1$ I $X_e = 0$.

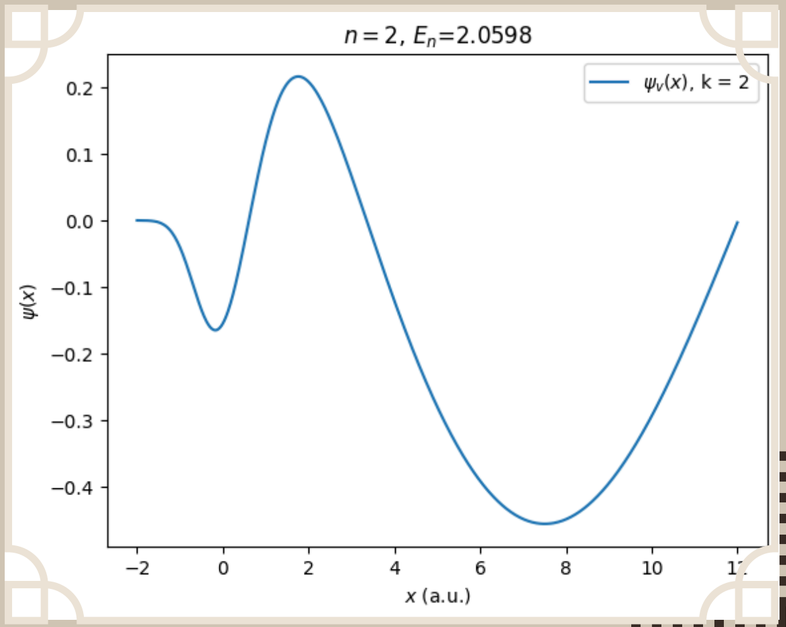
SUBSTITUINT ELS VALORS:

$V(X) = 2 \cdot [1 - EXP(-X)]^2$.



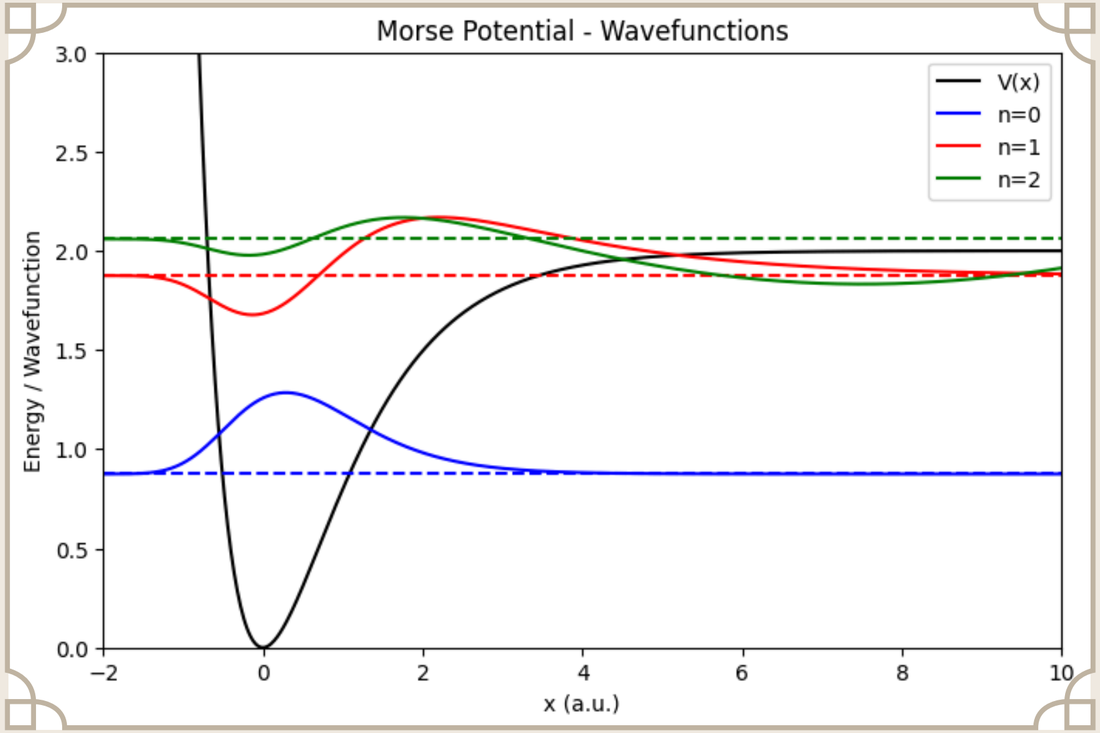


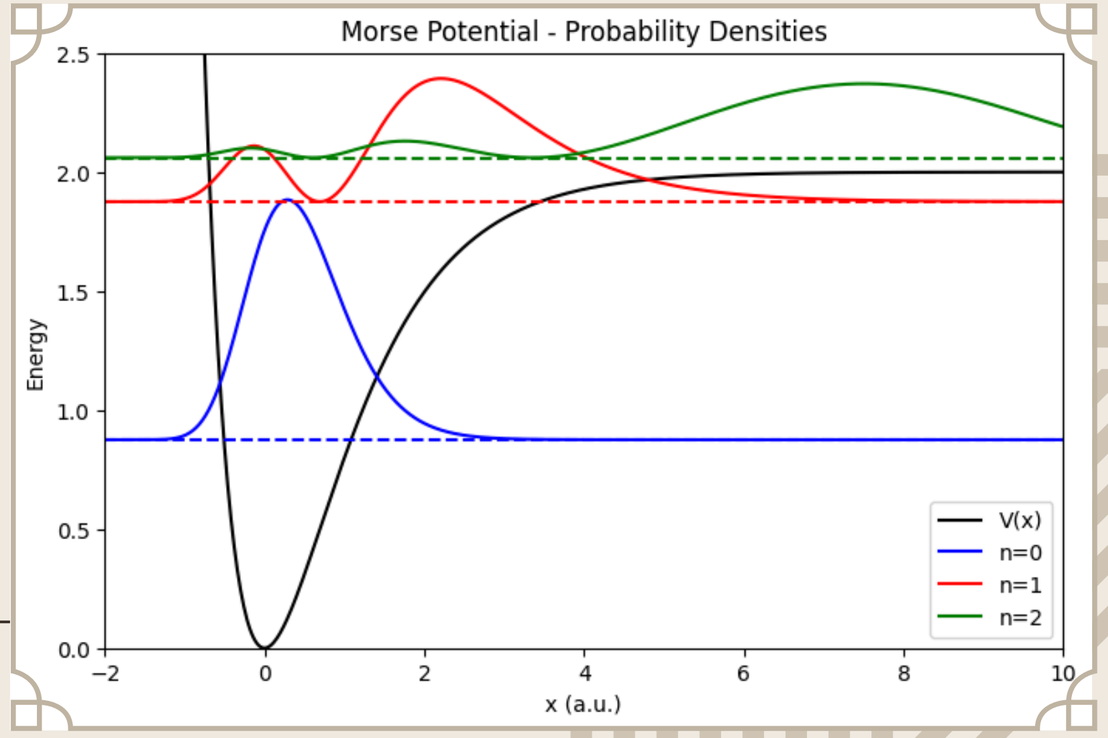




H=0,02
E(0)=0.8750
E(1)=1.8750
E(2)=2.0598
E(3)=2.2161
E(4)=2.4480

H=0,1
E(0)=0.8741
E(1)=1.8742
E(2)=2.0586
E(3)=2.2123
E(4)=2.4406


Morse Potential - Wavefunctions
Energy / Wavefunction
x (a.u.)
V(x)
n=0
n=1
n=2
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−2
0
2
4
6
8
10


Morse Potential - Probability Densities
Energy
x (a.u.)
V(x)
n=0
n=1
n=2
2.5
2.0
1.5
1.0
0.5
0.0
−2
0
2
4
6
8
10

MOLTES GRÀCIES